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VLMO: Unified Vision-Language Pre-Training with Mixture-of-Modality-Experts

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Abstract

We present a unified **V**ision-**L**anguage pretrained **Mo**del (**VLMO**) that jointly learns a dual encoder and a fusion encoder with a modular Transformer network. Specifically, we introduce **M**ixture-**o**f-**M**odality-**E**xperts (**MOME**) Transformer, where each block contains a pool of modality-specific experts and a shared self-attention layer. Because of the modeling flexibility of MOME, pretrained VLMO can be fine-tuned as a fusion encoder for vision-language classification tasks, or used as a dual encoder for efficient image-text retrieval. Moreover, we propose a stagewise pre-training strategy, which effectively leverages large-scale image-only and text-only data besides image-text pairs. Experimental results show that VLMO achieves state-of-the-art results on various vision-language tasks, including VQA, NLVR2 and image-text retrieval. The code and pretrained models are available at https://aka.ms/vlmo.

1 Introduction

Vision-Language (VL) pre-training [30, 41, 35, 26, 20, 23] learns generic cross-modal representations from large-scale image-text pairs. Previous models usually employ image-text matching, image-text contrastive learning, masked region classification/feature regression, word-region/patch alignment and masked language modeling to aggregate and align visual and linguistic information. Then the pretrained models can be directly fine-tuned on downstream vision-language tasks, such as VL retrieval and classification (visual question answering, visual reasoning, etc.).

Two mainstream architectures are widely used in previous work. CLIP [35] and ALIGN [18] adopt a *dual-encoder* architecture to encode images and text separately. Modality interaction is handled by the cosine similarity of the image and text feature vectors. The dual-encoder architecture is effective for retrieval tasks, especially for masses of images and text. Feature vectors of images and text can be pre-computed and stored. However, the shallow interaction between images and text is not enough to handle complex VL classification tasks. ViLT [20] finds that CLIP gives a relatively low accuracy on visual reasoning task. Another line of work [30, 41, 43, 3, 20, 23] relies on a fusion encoder with cross-modal attention to model image-text pairs. Multi-layer Transformer [45] networks are usually employed to fuse image and text representations. The *fusion-encoder* architecture achieves superior performance on VL classification tasks. But it requires to jointly encode all possible image-text pairs to compute similarity scores for retrieval tasks. The quadratic time complexity leads to a much slower inference speed than the dual-encoder models whose time complexity is linear.

In order to take advantage of the two types of architectures, we propose a unified **V**ision-**L**anguage pretrained **Mo**del (**VLMO**) that can be used as either a dual encoder to separately encode images and text for retrieval tasks, or used as a fusion encoder to model the deep interaction of image-text pairs for classification tasks. This is achieved by introducing **M**ixture-**o**f-**M**odality-**E**xperts

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VLMO：采用模态专家混合的统一视觉-语言预训练

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摘要

我们提出了一种统一的视觉-语言预训练模型（VLMO），通过模块化 Transformer 网络联合学习双编码器和融合编码器。具体而言，我们引入了模态专家混合（Mixture-of-Modality-Experts，MOME）Transformer，其中每个模块都包含一个模态特定专家池和一个共享的自注意力层。得益于 MOME 的建模灵活性，预训练的 VLMO 可以针对视觉-语言分类任务微调为融合编码器，也可以作为双编码器用于高效的图像-文本检索。此外，我们提出了一种分阶段预训练策略，除了图像-文本对之外，还能有效利用大规模的纯图像和纯文本数据。实验结果表明，VLMO 在包括 VQA、NLVR2 和图像-文本检索在内的多种视觉-语言任务上取得了业界领先的结果。代码和预训练模型可在 https://aka.ms/vlmo 获取。

1 引言

视觉-语言（VL）预训练 [30, 41, 35, 26, 20, 23] 从大规模图像-文本对中学习通用的跨模态表征。以往的模型通常采用图像-文本匹配、图像-文本对比学习、掩码区域分类/特征回归、词语-区域/图像块对齐以及掩码语言建模来聚合并对齐视觉与语言信息。随后，预训练模型可以直接在下游视觉-语言任务上进行微调，例如视觉-语言检索与分类（视觉问答、视觉推理等）。

以往工作中广泛采用两种主流架构。CLIP [35] 和 ALIGN [18] 采用 *dual-encoder* 架构分别对图像和文本进行编码。模态交互通过图像和文本特征向量的余弦相似度实现。双编码器架构对检索任务十分有效，尤其适用于海量图像和文本。图像和文本的特征向量可以预先计算并存储。然而，图像与文本之间浅层的交互不足以处理复杂的 VL 分类任务。ViLT [20] 发现，CLIP 在视觉推理任务上的准确率相对较低。另一类工作 [30, 41, 43, 3, 20, 23] 依赖具有跨模态注意力的融合编码器来建模图像-文本对。通常采用多层 Transformer [45] 网络来融合图像和文本表示。*fusion-encoder* 架构在 VL 分类任务上取得了更优的性能。但对于检索任务，它需要联合编码所有可能的图像-文本对来计算相似度分数。二次时间复杂度导致其推理速度远慢于时间复杂度为线性的双编码器模型。

为了充分利用这两种架构的优势，我们提出了一种统一的视觉-语言预训练模型（VLMO），既可以作为双编码器分别对图像和文本进行编码，以执行检索任务，也可以作为融合编码器，对图像-文本对之间的深层交互进行建模，以执行分类任务。这是通过引入模态专家混合模型实现的。

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(**MoME**) Transformer that can encode various modalities (images, text, and image-text pairs) within a Transformer block. MoME employs a pool of modality experts to replace the feed-forward network in standard Transformer. It captures modality-specific information by switching to different modality experts, and uses the shared self-attention across modalities to align visual and linguistic information. Specifically, MoME Transformer consists of three modality experts, namely vision expert for image encoding, language expert for text encoding, and vision-language expert for image-text fusion. Thanks to the modeling flexibility, we can reuse MoME Transformer with the shared parameters for different purposes, i.e., text-only encoder, image-only encoder, and image-text fusion encoder.

VLMo is jointly learned with three pre-training tasks, namely image-text contrastive learning, image-text matching, and masked language modeling. In addition, we propose a stagewise pre-training strategy to effectively leverage large-scale image-only and text-only corpus besides image-text pairs in VLMo pre-training. We first pretrain vision experts and self-attention modules of MoME Transformer on image-only data using masked image modeling proposed in BEiT [2]. We then pretrain language experts on text-only data using masked language modeling [10]. Finally, the model is used to initialize vision-language pre-training. By getting rid of the limited size of image-text pairs and their simple and short captions, stagewise pre-training on large amounts of image-only and text-only data helps VLMo to learn more generalizable representations.

Experimental results demonstrate that VLMo achieves state-of-the-art results on vision-language retrieval and classification tasks. Our model, used as a dual encoder, outperforms fusion-encoder-based models [3, 14, 20, 23] while enjoying a much faster inference speed on retrieval tasks. Moreover, our model also achieves state-of-the-art results on visual question answering (VQA) and natural language for visual reasoning (NLVR2), where VLMo is used as a fusion encoder.

Our main contributions are summarized as follows:

- We propose a unified vision-language pretrained model VLMo that can be used as a fusion encoder for classification tasks, or fine-tuned as a dual encoder for retrieval tasks.
- We introduce a general-purpose multimodal Transformer for vision-language tasks, namely MoME Transformer, to encode different modalities. It captures modality-specific information by modality experts, and aligns contents of different modalities by the self-attention module shared across modalities.
- We show that stagewise pre-training using large amounts of image-only and text-only data greatly improves our vision-language pretrained model.

2 Related Work

Pre-training with Transformer [45] backbone networks has substantially advanced the state of the art across natural language processing [34, 10, 28, 22, 11, 36, 1, 7, 8, 4–6, 31], computer vision [12, 44, 2] and vision-language [43, 41, 3, 49, 35, 18, 20, 23] tasks.

The approaches of vision-language pre-training can be divided into two categories. The first category utilizes a dual encoder to encode images and text separately, and uses cosine similarity or a linear projection layer to model the interaction between images and text [35, 18]. Image-text contrastive learning is usually employed to optimize the model. Dual-encoder models are effective for vision-language retrieval tasks. However, the simple interaction is not enough to handle tasks that require complex reasoning, such as visual reasoning and visual question answering (VL classification tasks). The second category models the interaction of images and text using a deep fusion encoder with cross-modal attention [43, 30, 41, 24, 51, 3, 26, 25, 14, 49, 16, 17, 20, 23, 46]. Image-text matching, masked language modeling, word-region/patch alignment, masked region classification and feature regression are widely used to train fusion-encoder-based models. These models achieve better performance for vision-language classification tasks, while the joint encoding of all image-text pairs leads to a slow inference speed for retrieval tasks. A large portion of fusion-encoder-based models rely on an off-the-shelf object detector like Faster R-CNN [37] to obtain image region features. Generating region features slows down the inference speed and renders the approach less scalable. Recently, Pixel-BERT [16] removes object detector and encodes images into grid features by convolutional neural networks. ALBEF [23] employs image Transformer [12, 44] to obtain the representations of images, and uses text Transformer [10] to learn the contextualized representations of text. These representations are then fused by cross-modal attention. ViLT [20] encodes images into

（MOME）Transformer 能够在一个 Transformer 块中对多种模态（图像、文本以及图像-文本对）进行编码。MOME 采用模态专家池来替代标准 Transformer 中的前馈网络。它通过切换至不同的模态专家来捕获特定于模态的信息，并利用跨模态共享的自注意力机制对齐视觉信息与语言信息。具体而言，MOME Transformer 由三个模态专家组成：用于图像编码的视觉专家、用于文本编码的语言专家，以及用于图像-文本融合的视觉-语言专家。得益于其建模灵活性，我们可以使用具有共享参数的 MOME Transformer 执行不同任务，即纯文本编码器、纯图像编码器和图像-文本融合编码器。

VLMO 通过三项预训练任务进行联合学习，即图像-文本对比学习、图像-文本匹配和掩码语言建模。此外，我们提出了一种分阶段预训练策略，以便在 VLMO 预训练过程中，除了使用图像-文本对之外，还能有效利用大规模的纯图像语料库和纯文本语料库。首先，我们使用 BEIT [2] 中提出的掩码图像建模，在纯图像数据上对 MOME Transformer 的视觉专家和自注意力模块进行预训练。随后，我们使用掩码语言建模 [10] 在纯文本数据上对语言专家进行预训练。最后，使用该模型初始化视觉-语言预训练。通过摆脱图像-文本对数量有限且其描述简单、简短的限制，在大量纯图像和纯文本数据上进行分阶段预训练，有助于 VLMO 学习更具泛化能力的表示。

实验结果表明，VLMO 在视觉-语言检索和分类任务上取得了最先进的结果。我们的模型作为双编码器使用时，在检索任务上的表现优于基于融合编码器的模型 [3, 14, 20, 23]，同时推理速度快得多。此外，在视觉问答（VQA）和视觉推理自然语言（NLVR2）任务上，我们的模型同样取得了最先进的结果，其中 VLMO 被用作融合编码器。

我们的主要贡献总结如下：

- 我们提出了一种统一的视觉-语言预训练模型 VLMO，可用作分类任务的融合编码器，或经过微调后用作检索任务的双编码器。
- 我们提出了一种用于视觉-语言任务的通用多模态 Transformer，即 MOME Transformer，用于编码不同模态。它通过模态专家捕获特定于模态的信息，并通过跨模态共享的自注意力模块对齐不同模态的内容。
- 我们表明，利用大量仅图像和仅文本数据进行分阶段预训练，能够显著提升我们的视觉语言预训练模型。

2 相关工作

使用 Transformer [45] 主干网络进行预训练，显著推动了自然语言处理 [34, 10, 28, 22, 11, 36, 1, 7, 8, 4–6, 31]、计算机视觉 [12, 44, 2] 以及视觉语言 [43, 41, 3, 49, 35, 18, 20, 23] 任务的发展。

视觉-语言预训练的方法可以分为两类。第一类使用双编码器分别对图像和文本进行编码，并利用余弦相似度或线性投影层来建模图像与文本之间的交互 [35, 18]。通常采用图像-文本对比学习来优化模型。双编码器模型在视觉-语言检索任务中表现有效。然而，这种简单的交互不足以处理需要复杂推理的任务，例如视觉推理和视觉问答（视觉-语言分类任务）。第二类使用带有跨模态注意力的深度融合编码器来建模图像与文本之间的交互 [43, 30, 41, 24, 51, 3, 26, 25, 14, 49, 16, 17, 20, 23, 46]。图像-文本匹配、掩码语言建模、词语-区域/图像块对齐、掩码区域分类和特征回归被广泛用于训练基于融合编码器的模型。这些模型在视觉-语言分类任务中取得了更好的性能，但对所有图像-文本对进行联合编码会导致检索任务的推理速度较慢。相当一部分基于融合编码器的模型依赖现成的目标检测器，例如 Faster R-CNN [37]，来获取图像区域特征。生成区域特征会降低推理速度，并使该方法的可扩展性变差。最近，Pixel-BERT [16] 移除了目标检测器，通过卷积神经网络将图像编码为网格特征。ALBEF [23] 使用图像 Transformer [12, 44] 获取图像的表示，并使用文本 Transformer [10] 学习文本的上下文表示。随后，这些表示通过跨模态注意力进行融合。ViLT [20] 将图像编码为

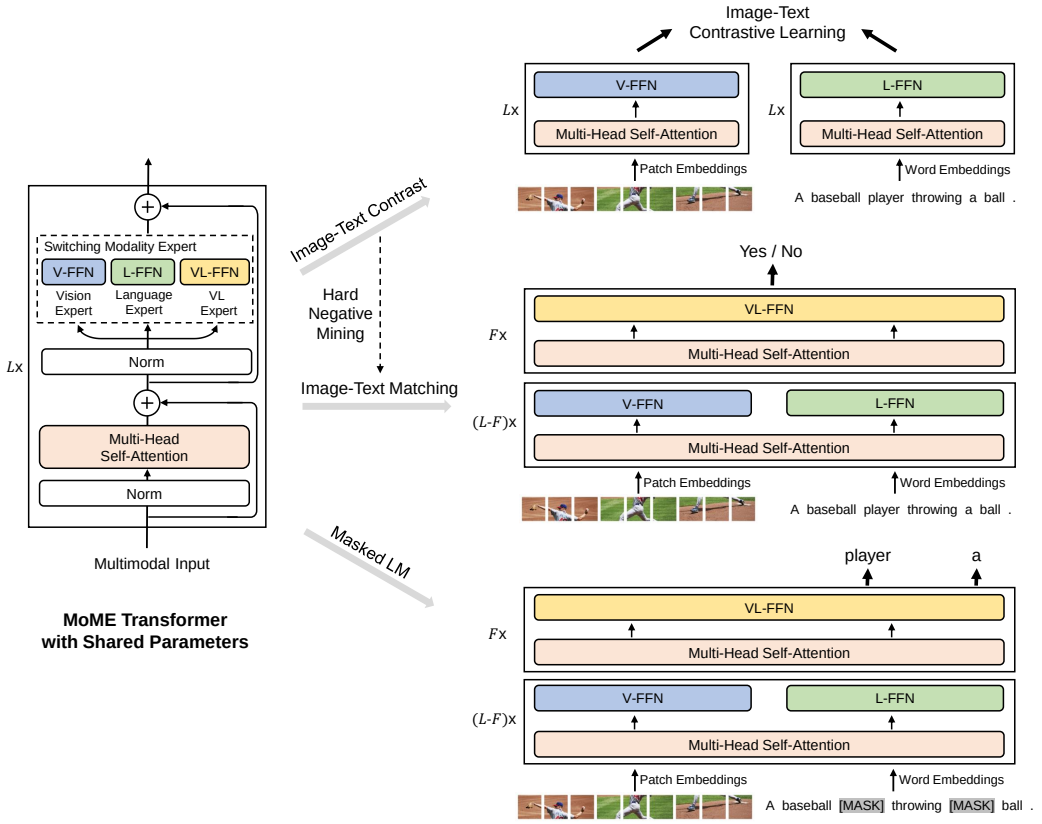


Figure 1: Overview of VLMO pre-training. We introduce mixture-of-modality-experts (MOME) Transformer to encode different modality input by modality-specific experts. The model parameters are shared across image-text contrastive learning, masked language modeling, and image-text matching pre-training tasks. During fine-tuning, the flexible modeling enables us to use VLMO as either a dual encoder (i.e., separately encode images and text for retrieval tasks) or a fusion encoder (i.e., jointly encode image-text pairs for better interaction across modalities).

patch embeddings, and then feed the concatenation of image patch embeddings and word embeddings into a Transformer network to learn contextualized representations and model the interaction of images and text.

Different from previous work, our unified pre-training using shared MOME Transformer enables the model perform separate encoding for retrieval tasks, and jointly encode image-text pairs to capture deeper interaction for classification tasks. Our model achieves competitive performance, while enjoying a faster inference speed for both retrieval and classification tasks.

3 Methods

Given image-text pairs, VLMO obtains image-only, text-only and image-text pair representations by the MOME Transformer network. As shown in Figure 1, the unified pre-training optimizes shared MOME Transformer with image-text contrastive learning on image-only and text-only representations, image-text matching and masked language modeling on image-text pair representations. Thanks to the modeling flexibility, the model can be used as a dual encoder for retrieval tasks to encode images and text separately during fine-tuning. It can also be fine-tuned as a fusion encoder to model deeper modality interaction of images and text for classification tasks.

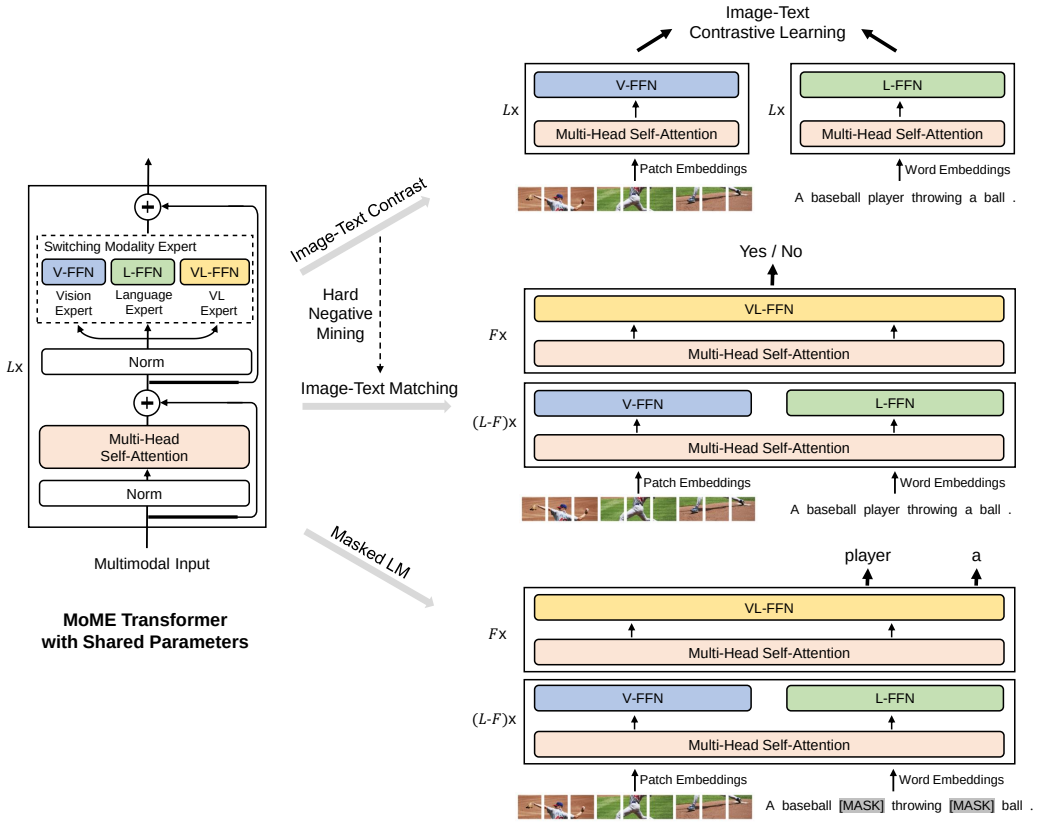


图 1：VLMO 预训练概览。我们引入混合模态专家（MOME）Transformer，通过模态特定专家对不同模态输入进行编码。模型参数在图文对比学习、掩码语言建模和图文匹配预训练任务之间共享。在微调期间，灵活的建模使我们能够将 VLMO 用作双编码器（即分别对图像和文本进行编码以用于检索任务）或融合编码器（即联合编码图文对以实现更好的跨模态交互）。

图像块嵌入，然后将图像块嵌入与词嵌入的拼接结果输入 Transformer 网络，以学习上下文表示并建模图像与文本之间的交互。

与以往工作不同，我们采用共享的 MOME Transformer 进行统一预训练，使模型能够针对检索任务进行独立编码，并对图像-文本对进行联合编码，以捕获更深层次的交互用于分类任务。我们的模型在检索和分类任务上都取得了具有竞争力的性能，同时享有更快的推理速度。

3 种方法

给定图像-文本对，VLMO 通过 MOME Transformer 网络获取仅图像、仅文本和图像-文本对表示。如图 1 所示，统一预训练在仅图像和仅文本表示上进行图像-文本对比学习，并在图像-文本对表示上进行图像-文本匹配和掩码语言建模，从而优化共享的 MOME Transformer 。得益于建模的灵活性，该模型在微调期间可用作双编码器，分别对图像和文本进行编码，以完成检索任务。它也可以作为融合编码器进行微调，以建模图像和文本之间更深层次的模态交互，从而完成分类任务。

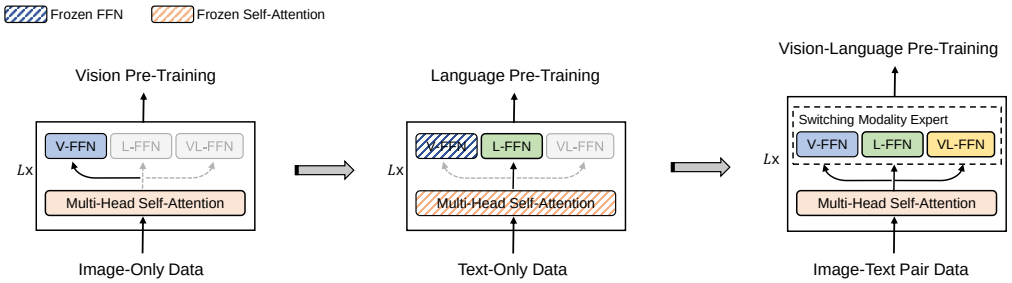


Figure 2: Stagewise pre-training using image-only and text-only corpora. We first pretrain the vision expert (V-FFN) and self-attention module on large-scale image-only data as in BEIT [2]. Then the parameters of vision expert and self-attention module are frozen, and we train the language expert (L-FFN) by masked language modeling on large amounts of text-only data. Finally, we train the whole model with vision-language pre-training.

3.1 Input Representations

Given an image-text pair, we encode the pair into image, text and image-text vector representations. These representations are then fed into the MOME Transformer to learn contextualized representations and align image and text feature vectors.

Image Representations Following vision Transformers [12, 44, 2], the 2D image $\boldsymbol{v} \in \mathbb{R}^{H \times W \times C}$ is split and reshaped into $N = HW/P^2$ patches $\boldsymbol{v}^p \in \mathbb{R}^{N \times (P^2 C)}$, where C is the number of channels, (H, W) is the resolution of the input image, and (P, P) is the patch resolution. The image patches are then flattened into vectors and are linearly projected to obtain patch embeddings. We also prepend a learnable special token `[I_CLS]` to the sequence. Finally, image input representations are obtained via summing patch embeddings, learnable 1D position embeddings $\boldsymbol{V}_{pos} \in \mathbb{R}^{(N+1) \times D}$ and image type embedding $\boldsymbol{V}_{type} \in \mathbb{R}^D$: $\boldsymbol{H}_0^v = [\boldsymbol{v}_{\texttt{[I_CLS]}}, \boldsymbol{V}\boldsymbol{v}_i^p, \ldots, \boldsymbol{V}\boldsymbol{v}_N^p] + \boldsymbol{V}_{pos} + \boldsymbol{V}_{type}$, where $\boldsymbol{H}_0^v \in \mathbb{R}^{(N+1) \times D}$, linear projection $\boldsymbol{V} \in \mathbb{R}^{(P^2 C) \times D}$.

Text Representations Following BERT [10], we tokenize the text to subword units by WordPiece [47]. A start-of-sequence token (`[T_CLS]`) and a special boundary token (`[T_SEP]`) are added to the text sequence. Text input representations $\boldsymbol{H}_0^w \in \mathbb{R}^{(M+2) \times D}$ are computed via summing the corresponding word embedding, text position embedding and text type embedding $\boldsymbol{H}_0^w = [\boldsymbol{w}_{\texttt{[T_CLS]}}, \boldsymbol{w}_i, \ldots, \boldsymbol{w}_M, \boldsymbol{w}_{\texttt{[T_SEP]}}] + \boldsymbol{T}_{pos} + \boldsymbol{T}_{type}$. M indicates the length of tokenized subword units.

Image-Text Representations We concatenate image and text input vectors to form the image-text input representations $\boldsymbol{H}_0^{vl} = [\boldsymbol{H}_0^w; \boldsymbol{H}_0^v]$

3.2 Mixture-of-Modality-Experts Transformer

Inspired by mixture-of-experts networks [40, 13], we propose a general-purpose multimodal Transformer for vision-language tasks, namely MOME Transformer, to encode different modalities. MOME Transformer introduces mixture of modality experts as a substitute of the feed forward network of standard Transformer. Given previous layer's output vectors $\boldsymbol{H}_{l-1}, l \in [1, L]$, each MOME Transformer block captures modality-specific information by switching to different modality expert, and employs multi-head self-attention (MSA) shared across modalities to align visual and linguistic contents. LN is short for layer normalization.

$$\boldsymbol{H}'_l = \text{MSA}(\text{LN}(\boldsymbol{H}_{l-1})) + \boldsymbol{H}_{l-1} \tag{1}$$

$$\boldsymbol{H}_l = \text{MoME-FFN}(\text{LN}(\boldsymbol{H}'_l)) + \boldsymbol{H}'_l \tag{2}$$

MoME-FFN selects an expert among multiple modality experts to process the input according to the modality of the input vectors $\boldsymbol{H}'_l$ and the index of the Transformer layer. Specifically, there are three modality experts: vision expert (V-FFN), language expert (L-FFN) and vision-language

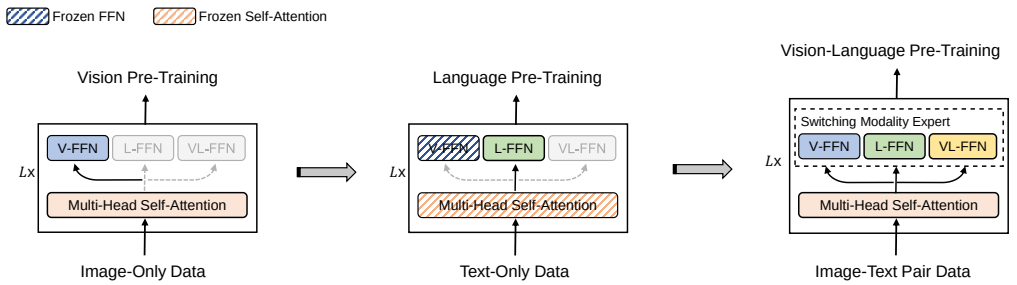


图 2：使用纯图像语料和纯文本语料进行分阶段预训练。我们首先按照 BEIT [2] 的方法，在大规模纯图像数据上预训练视觉专家（V-FFN）和自注意力模块。然后冻结视觉专家和自注意力模块的参数，并在大量纯文本数据上通过掩码语言建模训练语言专家（L-FFN）。最后，我们通过视觉-语言预训练训练整个模型。

3.1 输入表示形式

给定一个图像-文本对，我们将该对编码为图像、文本和图像-文本向量表示。然后将这些表示输入 MOME Transformer，以学习上下文化表示，并对齐图像和文本特征向量。

图像表示 遵循视觉 Transformer [12, 44, 2] 的做法，将 2D 图像 $\boldsymbol{v} \in \mathbb{R}^{H \times W \times C}$ 划分并重塑为 $N = HW/P^2$ 个图像块 $\boldsymbol{v}^p \in \mathbb{R}^{N \times (P^2 C)}$，其中 C 是通道数，(H, W) 是输入图像的分辨率，(P, P) 是图像块的分辨率。随后将图像块展平为向量，并通过线性投影得到图像块嵌入。我们还在序列前添加一个可学习的特殊标记 [I_CLS]。最后，将图像块嵌入、可学习的一维位置嵌入 $\boldsymbol{V}_{pos} \in \mathbb{R}^{(N+1) \times D}$ 和图像类型嵌入 $\boldsymbol{V}_{type} \in \mathbb{R}^{D}$ 相加，得到图像输入表示：$\boldsymbol{H}_0^v = [\boldsymbol{v}_{\texttt{[I_CLS]}}, \boldsymbol{V}\boldsymbol{v}_i^p, \ldots, \boldsymbol{V}\boldsymbol{v}_N^p] + \boldsymbol{V}_{pos} + \boldsymbol{V}_{type}$，其中 $\boldsymbol{H}_0^v \in \mathbb{R}^{(N+1) \times D}$，线性投影 $\boldsymbol{V} \in \mathbb{R}^{(P^2 C) \times D}$】【。

文本表示 遵循 BERT [10]，我们通过 WordPiece [47] 将文本分词为子词单元。序列起始标记（[T_CLS]）和特殊边界标记（[T_SEP]）被添加到文本序列中。文本输入表示 $\boldsymbol{H}_0^w \in \mathbb{R}^{(M+2) \times D}$ 通过将相应的词嵌入、文本位置嵌入和文本类型嵌入 $\boldsymbol{H}_0^w = [\boldsymbol{w}_{\texttt{[T_CLS]}}, \boldsymbol{w}_i, \ldots, \boldsymbol{w}_M, \boldsymbol{w}_{\texttt{[T_SEP]}}] + \boldsymbol{T}_{pos} + \boldsymbol{T}_{type}$ 相加来计算。M 表示分词后的子词单元长度。

图像-文本表示 我们将图像和文本输入向量拼接起来，形成图像-文本输入表示 $\boldsymbol{H}_0^{vl} = [\boldsymbol{H}_0^w; \boldsymbol{H}_0^v]$

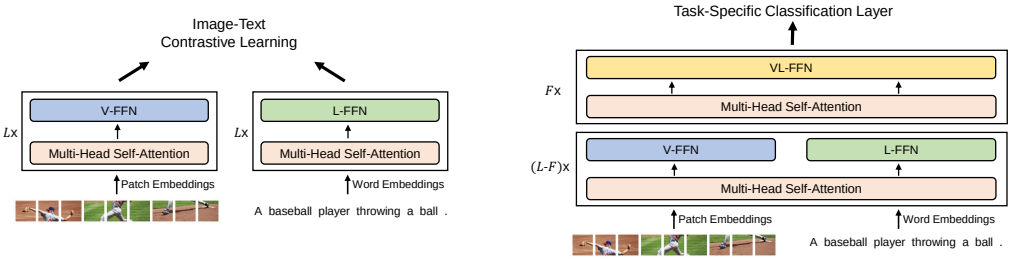
3.2 混合模态专家 Transformer

受混合专家网络 [40, 13] 的启发，我们提出了一种用于视觉-语言任务的通用多模态 Transformer，即 MOME Transformer，用于编码不同模态。MOME Transformer 引入了模态专家混合作为标准 Transformer 中前馈网络的替代方案。给定上一层的输出向量 $\boldsymbol{H}_{l-1}, l \in [1, L]$，每个 MOME Transformer 块通过切换到不同的模态专家来捕获模态特定信息，并采用跨模态共享的多头自注意力（MSA）来对齐视觉和语言内容。LN 是层归一化的缩写。

$$\boldsymbol{H}'_l = \text{MSA}(\text{LN}(\boldsymbol{H}_{l-1})) + \boldsymbol{H}_{l-1} \tag{1}$$

$$\boldsymbol{H}_l = \text{MoME-FFN}(\text{LN}(\boldsymbol{H}'_l)) + \boldsymbol{H}'_l \tag{2}$$

MoME-FFN 根据输入向量 $\boldsymbol{H}'_l$ 的模态以及 Transformer 层的索引，从多个模态专家中选择一个专家来处理输入。具体而言，共有三个模态专家：视觉专家（V-FFN）、语言专家（L-FFN）和视觉-语言



(a) Fine-Tuning on VL Retrieval Tasks

(b) Fine-Tuning on VL Classification Tasks

Figure 3: Fine-tuning VLMO on vision-language retrieval and classification tasks. The model can be fine-tuned as a dual encoder to separately encode image and text for retrieval tasks. VLMO can also be used as a fusion encoder to handle interaction of image-text pairs for classification tasks.

expert (VL-FFN). If the input is image-only or text-only vectors, we use vision expert for encoding images and language expert for encoding text. If the input consists of vectors of multiple modalities, such as the vectors of image-text pair, we employ vision expert and language expert to encode the respective modality vectors at the bottom Transformer layers. Vision-language expert is then used at the top layers to capture more modality interaction. Given the three types of input vectors, we obtain image-only, text-only and image-text contextualized representations.

3.3 Pre-Training Tasks

VLMO is jointly pretrained by image-text contrastive learning on the image and text representations, masked language modeling and image-text matching on the image-text pair representations with shared parameters.

Image-Text Contrast Given a batch of N image-text pairs, image-text contrastive learning aims to predict the matched pairs from $N \times N$ possible image-text pairs. There are $N^2 - N$ negative image-text pairs within a training batch.

The final output vectors of `[I_CLS]` token and `[T_CLS]` token are used as the aggregated representation of the image and text, respectively. Followed by a linear projection and normalization, we obtain image vectors $\{\hat{\boldsymbol{h}}_i^v\}_{i=1}^N$ and text vectors $\{\hat{\boldsymbol{h}}_i^w\}_{i=1}^N$ in a training batch to compute image-to-text and text-to-image similarities:

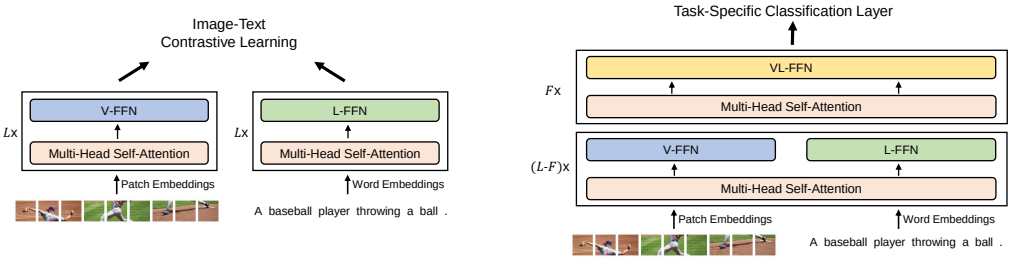
$$s_{i,j}^{i2t} = \hat{\boldsymbol{h}}_i^{v\intercal} \hat{\boldsymbol{h}}_j^w, \; s_{i,j}^{t2i} = \hat{\boldsymbol{h}}_i^{w\intercal} \hat{\boldsymbol{h}}_j^v \tag{3}$$

$$p_i^{i2t} = \frac{\exp(s_{i,i}^{i2t}/\sigma)}{\sum_{j=1}^N \exp(s_{i,j}^{i2t}/\sigma)}, \; p_i^{t2i} = \frac{\exp(s_{i,i}^{t2i}/\sigma)}{\sum_{j=1}^N \exp(s_{i,j}^{t2i}/\sigma)} \tag{4}$$

Where $s_{i,j}^{i2t}$ represents image-to-text similarity of image of i-th pair and text of j-th pair, $s_{i,j}^{t2i}$ is the text-to-image similarity. $\hat{\boldsymbol{h}}_i^w \in \mathbb{R}^D$ and $\hat{\boldsymbol{h}}_j^v \in \mathbb{R}^D$ indicate the normalized vectors of i-th text and j-th image, σ is a learned temperature parameter. p_i^{i2t} and p_i^{t2i} are the $\mathrm{softmax}$-normalized similarities. Cross-entropy losses over image-to-text and text-to-image similarities are used to train the model.

Masked Language Modeling Following BERT [10], we randomly choose tokens in the text sequence, and replace them with the `[MASK]` token. The model is trained to predict these masked tokens from all the other unmasked tokens and vision clues. We use 15% masking probability as in BERT. The final output vectors of masked tokens are fed into a classifier over the whole text vocabulary with cross-entropy loss.

Image-Text Matching Image-text matching aims to predict whether the image and text is matched. We use the final hidden vector of the `[T_CLS]` token to represent the image-text pair, and feed the vector into a classifier with cross-entropy loss for binary classification. Inspired by ALBEF [23], we sample hard negative image-text pairs based on the contrastive image-to-text and text-to-image



(a) Fine-Tuning on VL Retrieval Tasks

(b) Fine-Tuning on VL Classification Tasks

图 3：在视觉-语言检索和分类任务上对 VLMO 进行微调。该模型可以作为双编码器进行微调，分别对图像和文本进行编码，以完成检索任务。VLMO 也可以作为融合编码器，用于处理图像-文本对之间的交互，以完成分类任务。

专家（VL-FFN）。如果输入仅包含图像向量或仅包含文本向量，我们使用视觉专家对图像进行编码，并使用语言专家对文本进行编码。如果输入由多种模态的向量组成，例如图像-文本对的向量，我们在底部 Transformer 层使用视觉专家和语言专家分别对相应模态的向量进行编码。随后，在顶部层使用视觉-语言专家，以捕获更多的模态交互。对于这三类输入向量，我们可以获得仅图像、仅文本以及图像-文本的上下文化表示。

3.3 预训练任务

VLMO通过在图像和文本表示上进行图像-文本对比学习，以及在图像-文本对表示上进行掩码语言建模和图像-文本匹配，利用共享参数进行联合预训练。

图像-文本对比 给定一批 N 个图像-文本对，图像-文本对比学习旨在从 $N \times N$ 个可能的图像-文本对中预测匹配的配对。一个训练批次中有 $N^2 - N$ 个负图像-文本对。

[I_CLS] 标记和 [T_CLS] 标记的最终输出向量分别作为图像和文本的聚合表示。经过线性投影和归一化后，我们获得训练批次中的图像向量 $\{\hat{\boldsymbol{h}}_i^v\}_{i=1}^N$ 和文本向量 $\{\hat{\boldsymbol{h}}_i^w\}_{i=1}^N$，以计算图像到文本和文本到图像的相似度：

$$s_{i,j}^{i2t} = \hat{\boldsymbol{h}}_i^{v\intercal} \hat{\boldsymbol{h}}_j^w, \; s_{i,j}^{t2i} = \hat{\boldsymbol{h}}_i^{w\intercal} \hat{\boldsymbol{h}}_j^v \tag{3}$$

$$p_i^{i2t} = \frac{\exp(s_{i,i}^{i2t}/\sigma)}{\sum_{j=1}^N \exp(s_{i,j}^{i2t}/\sigma)}, \; p_i^{t2i} = \frac{\exp(s_{i,i}^{t2i}/\sigma)}{\sum_{j=1}^N \exp(s_{i,j}^{t2i}/\sigma)} \tag{4}$$

其中，$s_{i,j}^{i2t}$ 表示第 i 对图像与第 j 对文本之间的图像到文本相似度，$s_{i,j}^{t2i}$ 表示文本到图像相似度。$\hat{\boldsymbol{h}}_i^w \in \mathbb{R}^D$ 和 $\hat{\boldsymbol{h}}_j^v \in \mathbb{R}^D$ 分别表示第 i 个文本和第 j 个图像的归一化向量，σ 是一个学习得到的温度参数。p_i^{i2t} 和 p_i^{t2i} 是经过 softmax 归一化的相似度。使用图像到文本和文本到图像相似度上的交叉熵损失来训练模型。

掩码语言建模 遵循BERT [10] 的方法，我们随机选择文本序列中的词元，并将其替换为 [MASK] 词元。模型经过训练，可以根据其他所有未被掩码的词元和视觉线索来预测这些被掩码的词元。我们采用与 BERT 相同的 15% 掩码概率。被掩码词元的最终输出向量被输入一个覆盖整个文本词汇表的分类器，并使用交叉熵损失进行训练。

图像-文本匹配 图像-文本匹配旨在预测图像与文本是否匹配。我们使用 [T_CLS] 标记的最终隐藏向量来表示图像-文本对，并将该向量输入分类器，通过交叉熵损失进行二分类。受 ALBEF [23] 的启发，我们基于图像到文本和文本到图像的对比学习来采样难负图像-文本对。

similarities. Different from ALBEF [23], which samples hard negatives from training examples of the single GPU (we named it as local hard negative mining). We propose global hard negative mining and sample hard negative image-text pairs from more training examples gathered from all GPUs. Global hard negative mining can find more informative image-text pairs and significantly improves our model.

3.4 Stagewise Pre-Training

We introduce a stagewise pre-training strategy, which leverages large-scale image-only and text-only corpus to improve the vision-language model. As present in Figure 2, we first perform vision pre-training on image-only data, and then perform language pre-training on text-only data to learn general image and text representations. The model is used to initialize the vision-language pre-training to learn the alignment of visual and linguistic information. For vision pre-training, we train the attention module and vision expert of MoME Transformer as in BEiT [2] on image-only data. We directly utilize the pretrained parameters of BEiT to initialize the attention module and vision expert. For language pre-training, we freeze parameters of the attention module and vision expert, and utilize masked language modeling [10] to optimize the language expert on text-only data. Compared with image-text pairs, image-only and text-only data are easier to collect. In addition, text data of image-text pairs is usually short and simple. Pre-training on image-only and text-only corpus improves the generalization on complex pairs.

3.5 Fine-Tuning VLMo on Downstream Tasks

As present in Figure 3, our model can be fine-tuned to adapt to various vision-language retrieval and classification tasks.

Vision-Language Classification For classification tasks such as visual question answering and visual reasoning, VLMo is used as a fusion encoder to model modality interaction of images and text. We use the final encoding vector of the token `[T_CLS]` as the representation of the image-text pair, and feed it to a task-specific classifier layer to predict the label.

Vision-Language Retrieval For retrieval tasks, VLMo can be used as a dual encoder to encode images and text separately. During fine-tuning, our model is optimized for the image-text contrastive loss. During inference, we compute representations of all images and text, and then use dot product to obtain image-to-text and text-to-image similarity scores of all possible image-text pairs. Separate encoding enables a much faster inference speed than fusion-encoder-based models.

4 Experiments

We pretrain our model using large-scale image-text pairs and evaluate the model on visual-linguistic classification and retrieval tasks.

4.1 Pre-Training Setup

Following previous work [3, 20], our pre-training data consists of four image captioning datasets: Conceptual Captions (CC) [39], SBU Captions [32], COCO [27] and Visual Genome (VG) [21] datasets. There are about 4M images and 10M image-text pairs in the pre-training data.

Our models adopt the same network configuration as ViT [12] and BEiT [2]. VLMo-Base consists of 12-layer Transformer blocks with 768 hidden size and 12 attention heads. VLMo-Large is a 24-layer Transformer network with 1024 hidden size and 16 attention heads. The intermediate size of feed-forward networks is 3072 and 4096 for base-size and large-size models, respectively. VLMo-Base uses vision-language expert on the top two Transformer layers, and VLMo-Large introduces vision-language expert on the top three layers. For images, the input resolution is 224×224 and the patch size is 16×16 during pre-training. We apply RandAugment [9] to the input images. The tokenizer of the uncased version of BERT is employed to tokenize the text. The maximum text sequence length is set to 40. We also employ whole word masking for the masked language modeling pre-training task. We pretrain the models for 200k steps with 1024 batch size. We utilize AdamW [29] optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.98$. The peak learning is 2e-4 for the base-size model,

相似性。不同于 ALBEF [23]，后者从单个 GPU 的训练样本中采样困难负样本（我们称之为局部困难负样本挖掘），我们提出了全局困难负样本挖掘，从所有 GPU 汇集的更多训练样本中采样困难的图像-文本对。全局困难负样本挖掘能够找到信息量更大的图像-文本对，并显著提升我们的模型。

3.4 分阶段预训练

我们引入了一种分阶段预训练策略，利用大规模的纯图像语料库和纯文本语料库来提升视觉语言模型。如图 2 所示，我们首先在纯图像数据上进行视觉预训练，然后在纯文本数据上进行语言预训练，以学习通用的图像和文本表征。随后，使用该模型初始化视觉语言预训练，以学习视觉信息与语言信息之间的对齐关系。对于视觉预训练，我们按照 BEIT [2] 的方式，在纯图像数据上训练 MOME Transformer 的注意力模块和视觉专家。我们直接利用 BEIT 的预训练参数来初始化注意力模块和视觉专家。对于语言预训练，我们冻结注意力模块和视觉专家的参数，并利用掩码语言建模 [10] 在纯文本数据上优化语言专家。与图文对相比，纯图像数据和纯文本数据更易于收集。此外，图文对中的文本通常较短且简单。在纯图像语料库和纯文本语料库上进行预训练，可以提升模型在复杂图文对上的泛化能力。

3.5 在下游任务上对 VLMO 进行微调

如图3所示，我们的模型可以通过微调来适应各种视觉-语言检索和分类任务。

视觉-语言分类 对于视觉问答和视觉推理等分类任务，VLMO 被用作融合编码器来建模图像与文本之间的模态交互。我们使用标记 [T_CLS] 的最终编码向量作为图像-文本对的表示，并将其输入到特定任务的分类器层中以预测标签。

视觉-语言检索　　对于检索任务，VLMO 可以作为双编码器，分别对图像和文本进行编码。在微调过程中，我们的模型针对图像-文本对比损失进行优化。在推理过程中，我们计算所有图像和文本的表示，然后使用点积获得所有可能的图像-文本对的图像到文本和文本到图像相似度分数。与基于融合编码器的模型相比，独立编码能够实现更快的推理速度。

4 实验

我们使用大规模图像-文本对对模型进行预训练，并在视觉语言分类和检索任务上对模型进行评估。

4.1 预训练设置

遵循先前的工作 [3, 20]，我们的预训练数据由四个图像描述数据集组成：Conceptual Captions（CC）[39]、SBU Captions [32]、COCO [27] 和 Visual Genome（VG）[21] 数据集。预训练数据中约有 400 万张图像和 1000 万个图像-文本对。

我们的模型采用与 ViT [12] 和 BEIT [2] 相同的网络配置。VLMO-Base 由 12 层 Transformer 模块组成，隐藏层大小为 768，包含 12 个注意力头。VLMO-Large 是一个 24 层的 Transformer 网络，隐藏层大小为 1024，包含 16 个注意力头。对于基础规模和大型模型，前馈网络的中间层大小分别为 3072 和 4096。VLMO-Base 在顶部两个 Transformer 层中使用视觉-语言专家模块，而 VLMO-Large 则在顶部三个层中引入视觉-语言专家模块。对于图像，在预训练期间，输入分辨率为 224×224，图像块大小为 16×16。我们对输入图像应用 RandAugment [9]。文本标记化采用 BERT 的不区分大小写版本的分词器。文本序列的最大长度设置为 40。对于掩码语言建模预训练任务，我们还采用整词掩码。我们使用批量大小为 1024 对模型进行 200k 步预训练。我们采用 AdamW [29] 优化器，使用 $\beta_1 = 0.9$、$\beta_2 = 0.98$。基础规模模型的峰值学习率为 2e-4，

Model	# Pretrain Images	VQA		NLVR2	
		test-dev	test-std	dev	test-P
Base-Size Models Pretrained on COCO, VG, SBU and CC datasets					
UNITER-Base [3]	4M	72.70	72.91	77.18	77.85
VILLA-Base [14]	4M	73.59	73.67	78.39	79.30
UNIMO-Base [25]	4M	73.79	74.02	-	-
ViLT-Base [20]	4M	71.26	-	75.70	76.13
ALBEF-Base [23]	4M	74.54	74.70	80.24	80.50
VLMO-Base	4M	**76.64**	**76.89**	**82.77**	**83.34**
Large-Size Models Pretrained on COCO, VG, SBU and CC datasets					
UNITER-Large [3]	4M	73.82	74.02	79.12	79.98
VILLA-Large [14]	4M	74.69	74.87	79.76	81.47
UNIMO-Large [25]	4M	75.06	75.27	-	-
VLMO-Large	4M	**79.94**	**79.98**	**85.64**	**86.86**
Models Pretrained on More Data					
VinVL-Large [49]	5.7M	76.52	76.60	82.67	83.98
SimVLM-Large [46]	1.8B	79.32	79.56	84.13	84.84
SimVLM-Huge [46]	1.8B	80.03	80.34	84.53	85.15
Florence-Huge [48]	900M	80.16	80.36	-	-
VLMO-Large++	1.0B	**82.88**	**82.78**	**88.62**	**89.54**

Table 1: Fine-tuning results of base-size and large-size VLMO on vision-language classification datasets. VLMO-Large++ is the model trained on one billion noisy image-text pairs with a larger batch size. We report vqa-score on VQA test-dev and test-standard split, and report accuracy for NLVR2 development and public test set (test-P).

5e-5 for the large-size model. Weight decay is set to 0.01. We use linear warmup over the first 2.5k steps and linear decay. The vision-language pre-training of base-size model takes about two days using 64 Nvidia Tesla V100 32GB GPU cards, and the large-size model takes about three days using 128 Nvidia Tesla V100 32GB GPU cards.

4.2 Training on Larger-scale Datasets

We scale up vision-language representation learning by training VLMO-Large on one billion noisy web image-text pairs with a larger batch size. We first pretrain the model for 200k steps with 16k batch size, and then continue train the model for 100k steps with 32k batch size. The other hyper-parameters are the same as the training on 4M data. Please refer to the supplementary material for more details of hyper-parameters used for pre-training and fine-tuning.

4.3 Evaluation on Vision-Language Classification Tasks

We first conduct fine-tuning experiments on two widely used classification datasets: visual question answering [15] and natural language for visual reasoning [42]. The model is fine-tuned as a fusion encoder to model deeper interaction.

Visual Question Answering (VQA) For VQA, a natural image and a question are given, the task is to generate/choose the correct answer. We train and evaluate the model on VQA 2.0 dataset [15]. Following common practices, we convert VQA 2.0 to a classification task, and choose the answer from a shared set consists of $3,129$ answers. We use the final encoding vector of the `[T_CLS]` token as the representation of the image-question pair and feed it to a classifier layer to predict the answer.

Natural Language for Visual Reasoning (NLVR2) The NLVR2 [42] dataset requires the model to predict whether a text description is true about a pair of images. Following OSCAR [26] and VinVL [49], we convert the triplet input to two image-text pairs, each containing the text description and one image. We concatenate the final output vectors of the `[T_CLS]` token of the two input pairs. The concatenated vector is then fed into a classification layer to predict the label.

Model	# Pretrain Images	VQA		NLVR2	
		test-dev	test-std	dev	test-P
Base-Size Models Pretrained on COCO, VG, SBU and CC datasets					
UNITER-Base [3]	4M	72.70	72.91	77.18	77.85
VILLA-Base [14]	4M	73.59	73.67	78.39	79.30
UNIMO-Base [25]	4M	73.79	74.02	-	-
ViLT-Base [20]	4M	71.26	-	75.70	76.13
ALBEF-Base [23]	4M	74.54	74.70	80.24	80.50
VLMo-Base	4M	**76.64**	**76.89**	**82.77**	**83.34**
Large-Size Models Pretrained on COCO, VG, SBU and CC datasets					
UNITER-Large [3]	4M	73.82	74.02	79.12	79.98
VILLA-Large [14]	4M	74.69	74.87	79.76	81.47
UNIMO-Large [25]	4M	75.06	75.27	-	-
VLMo-Large	4M	**79.94**	**79.98**	**85.64**	**86.86**
Models Pretrained on More Data					
VinVL-Large [49]	5.7M	76.52	76.60	82.67	83.98
SimVLM-Large [46]	1.8B	79.32	79.56	84.13	84.84
SimVLM-Huge [46]	1.8B	80.03	80.34	84.53	85.15
Florence-Huge [48]	900M	80.16	80.36	-	-
VLMo-Large++	1.0B	**82.88**	**82.78**	**88.62**	**89.54**

表 1：VLMO-base 和 VLMO-large 在视觉-语言分类数据集上的微调结果。VLMO-Large++ 是在十亿个带噪声的图像-文本对上以更大的批量大小训练的模型。我们报告了 VQA test-dev 和 test-standard 划分上的 vqa-score，以及 NLVR2 development 和公开测试集（test-P）上的准确率。

大型模型为 5e-5。权重衰减设置为 0.01。我们在前 2.5k 个步骤中采用线性预热，然后进行线性衰减。基础尺寸模型的视觉语言预训练使用 64 张 Nvidia Tesla V100 32GB GPU 卡，耗时约两天；大型模型使用 128 张 Nvidia Tesla V100 32GB GPU 卡，耗时约三天。

4.2 在更大规模数据集上训练

我们通过使用更大的批量大小，在十亿个有噪声的网页图像-文本对上训练 VLMO-Large，来扩展视觉-语言表征学习。我们首先以 16k 的批量大小对模型预训练 200k 步，然后继续以 32k 的批量大小训练模型 100k 步。其他超参数与在 4M 数据上的训练相同。有关预训练和微调所用超参数的更多细节，请参阅补充材料。

4.3 视觉-语言分类任务上的评估

我们首先在两个广泛使用的分类数据集上进行微调实验：视觉问答 [15] 和用于视觉推理的自然语言 [42]。该模型被微调为融合编码器，以建模更深层次的交互。

视觉问答（VQA）对于 VQA，给定一张自然图像和一个问题，任务是生成或选择正确的答案。我们在 VQA 2.0 数据集 [15] 上训练并评估模型。遵循常见做法，我们将 VQA 2.0 转换为分类任务，并从由 3, 129 个答案组成的共享集合中选择答案。我们使用 [T_CLS] 标记的最终编码向量作为图像-问题对的表示，并将其输入分类器层以预测答案。

视觉推理的自然语言（NLVR2）NLVR2 [42] 数据集要求模型预测文本描述对于一对图像是否为真。按照 OSCAR [26] 和 VinVL [49] 的做法，我们将三元组输入转换为两个图像-文本对，每个图像-文本对都包含文本描述和一张图像。我们将两个输入对的 [T_CLS] 标记的最终输出向量进行拼接。然后将拼接后的向量输入分类层以预测标签。

Model	# Pretrain Images	MSCOCO (5K test set)						Flickr30K (1K test set)					
		TR			IR			TR			IR		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
Base-Size Models Pretrained on COCO, VG, SBU and CC datasets													
UNITER-Base	4M	64.4	87.4	93.1	50.3	78.5	87.2	85.9	97.1	98.8	72.5	92.4	96.1
VILLA-Base	4M	-	-	-	-	-	-	86.6	97.9	99.2	74.7	92.9	95.8
ViLT-Base	4M	61.5	86.3	92.7	42.7	72.9	83.1	83.5	96.7	98.6	64.4	88.7	93.8
ALBEF-Base‡	4M	73.1	91.4	96.0	56.8	81.5	89.2	**94.3**	**99.4**	99.8	**82.8**	**96.7**	**98.4**
VLMo-Base†	4M	**74.8**	**93.1**	**96.9**	**57.2**	**82.6**	**89.8**	92.3	**99.4**	**99.9**	79.3	95.7	97.8
Large-Size Models Pretrained on COCO, VG, SBU and CC datasets													
UNITER-Large	4M	65.7	88.6	93.8	52.9	79.9	88.0	87.3	98.0	99.2	75.6	94.1	96.8
VILLA-Large	4M	-	-	-	-	-	-	87.9	97.5	98.8	76.3	94.2	96.8
VLMo-Large†	4M	**78.2**	**94.4**	**97.4**	**60.6**	**84.4**	**91.0**	**95.3**	**99.9**	**100.0**	**84.5**	**97.3**	**98.6**
Models Pretrained on More Data													
VinVL-Large	5.7M	75.4	92.9	96.2	58.8	83.5	90.3	-	-	-	-	-	-
ALIGN-Large†	1.8B	77.0	93.5	96.9	59.9	83.3	89.8	95.3	99.8	**100.0**	84.9	97.4	98.6
Florence-Huge†	900M	81.8	95.2	-	63.2	85.7	-	**97.2**	99.9	-	87.9	98.1	-
VLMo-Large++†	1.0B	**83.1**	**96.0**	**98.2**	**65.2**	**86.5**	**92.2**	96.8	**100.0**	**100.0**	**88.1**	**98.4**	**99.3**

Table 2: Fine-tuning results of text-retrieval (TR) and image-retrieval (IR) on COCO and Flickr30K. †: ALIGN, Florence and our model encode images and text separately, and then employ a shallow interaction (dot product) to obtain the similarity scores. ‡: ALBEF first encodes images and text separately to obtain the top-k candidates, and then feed these representations into a fusion encoder to rerank the candidates. The others require to encode all image-text combinations by a fusion encoder. VLMo-Large++ represents the model trained on one billion noisy image-text pairs with a larger batch size.

We present the results of VL classification tasks in Table 1. VLMo achieves state-of-the-art performance and substantially outperforms previous methods. Our large-size model even outperforms SimVLM-Huge [46] and Florence-Huge [48] by a large margin, which consists of more parameters and are also trained on larger-scale image-text pairs. Our model uses a simple linear projection to embed images as in ViLT [20]. This leads to a significant speedup compared with previous models using image region features, which are extracted by an off-the-shelf object detector [30, 41, 3, 14, 25, 49].

4.4 Evaluation on Vision-Language Retrieval Tasks

The retrieval tasks contain image-to-text retrieval and text-to-image retrieval. We evaluate the model on the widely used COCO [27] and Flickr30K [33] datasets, and use the Karpathy split [19] for both datasets. The model is used as a dual encoder for retrieval tasks. We encode images and text separately and compute their similarity scores by the dot product of image and text vectors.

As present in Table 2, VLMo achieves competitive performance with previous fusion-encoder-based models while having a much faster speed. Fusion-encoder-based models need to jointly encode all possible image-text pairs to compute their similarity scores, which requires quadratic time complexity. Moreover, our large-size model even outperforms the huge-size model of Florence [48], which also trained on massive image-text pairs using a larger batch size. VLMo pre-training can effectively leverage larger-scale noisy pairs and benefit from large batch training.

4.5 Evaluation on Vision Tasks

As shown in Table 3, we use VLMo as an image-only encoder and evaluate it on image classification (ImageNet [38]) and semantic segmentation (ADE20K [50]) tasks. The model also achieves competitive performance, even slightly better than the BEiT model used for the initialization of VLMo. The image resolution is 224×224 for ImageNet, and 512×512 for ADE20K. We perform intermediate fine-tuning [2] on ImageNet-21k for all three models.

Model	# Pretrain Images	MSCOCO (5K test set)						Flickr30K (1K test set)					
		TR			IR			TR			IR		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
Base-Size Models Pretrained on COCO, VG, SBU and CC datasets													
UNITER-Base	4M	64.4	87.4	93.1	50.3	78.5	87.2	85.9	97.1	98.8	72.5	92.4	96.1
VILLA-Base	4M	-	-	-	-	-	-	86.6	97.9	99.2	74.7	92.9	95.8
ViLT-Base	4M	61.5	86.3	92.7	42.7	72.9	83.1	83.5	96.7	98.6	64.4	88.7	93.8
ALBEF-Base‡	4M	73.1	91.4	96.0	56.8	81.5	89.2	**94.3**	**99.4**	99.8	**82.8**	**96.7**	**98.4**
VLMo-Base†	4M	**74.8**	**93.1**	**96.9**	**57.2**	**82.6**	**89.8**	92.3	**99.4**	**99.9**	79.3	95.7	97.8
Large-Size Models Pretrained on COCO, VG, SBU and CC datasets													
UNITER-Large	4M	65.7	88.6	93.8	52.9	79.9	88.0	87.3	98.0	99.2	75.6	94.1	96.8
VILLA-Large	4M	-	-	-	-	-	-	87.9	97.5	98.8	76.3	94.2	96.8
VLMo-Large†	4M	**78.2**	**94.4**	**97.4**	**60.6**	**84.4**	**91.0**	**95.3**	**99.9**	**100.0**	**84.5**	**97.3**	**98.6**
Models Pretrained on More Data													
VinVL-Large	5.7M	75.4	92.9	96.2	58.8	83.5	90.3	-	-	-	-	-	-
ALIGN-Large†	1.8B	77.0	93.5	96.9	59.9	83.3	89.8	95.3	99.8	**100.0**	84.9	97.4	98.6
Florence-Huge†	900M	81.8	95.2	-	63.2	85.7	-	**97.2**	99.9	-	87.9	98.1	-
VLMo-Large++†	1.0B	**83.1**	**96.0**	**98.2**	**65.2**	**86.5**	**92.2**	96.8	**100.0**	**100.0**	**88.1**	**98.4**	**99.3**

表2：在 COCO 和 Flickr30K 上进行文本检索（TR）和图像检索（IR）的微调结果。†：ALIGN、Florence 以及我们的模型分别对图像和文本进行编码，然后采用浅层交互（点积）来获得相似度分数。‡：ALBEF 首先分别对图像和文本进行编码，以获得 top-k 候选项，然后将这些表示输入融合编码器以对候选项重新排序。其他方法需要通过融合编码器对所有图像-文本组合进行编码。VLMO-Large++ 表示使用更大批量大小、在十亿个有噪声图像-文本对上训练的模型。

我们在表 1 中展示了 VL 分类任务的结果。VLMO 达到了最先进的性能，并显著优于以往的方法。我们的大型模型甚至以较大幅度超过了 SimVLM-Huge [46] 和 Florence-Huge [48]，后两者拥有更多参数，并且也是在规模更大的图像-文本对上进行训练的。与 ViLT [20] 一样，我们的模型使用简单的线性投影来嵌入图像。与以往使用图像区域特征的模型相比，这带来了显著的加速；这些图像区域特征由现成的目标检测器 [30, 41, 3, 14, 25, 49] 提取。

4.4 视觉-语言检索任务评估

检索任务包括图像到文本检索和文本到图像检索。我们在广泛使用的 COCO [27] 和 Flickr30K [33] 数据集上评估模型，并对两个数据集均采用 Karpathy 划分 [19]。在检索任务中，模型被用作双编码器。我们分别对图像和文本进行编码，并通过图像向量与文本向量的点积计算它们的相似度分数。

如表 2 所示，VLMO 在具备更快速度的同时，取得了与以往基于融合编码器的模型相竞争的性能。基于融合编码器的模型需要联合编码所有可能的图像-文本对以计算其相似度分数，这需要二次时间复杂度。此外，我们的大尺寸模型甚至优于 Florence [48] 的超大尺寸模型，后者同样使用更大的批量大小在海量图像-文本对上进行训练。VLMO 预训练能够有效利用更大规模的噪声配对，并受益于大批量训练。

4.5 视觉任务评估

如表 3 所示，我们将 VLMO 用作仅图像编码器，并在图像分类（ImageNet [38]）和语义分割（ADE20K [50]）任务上对其进行评估。该模型也取得了有竞争力的性能，甚至略优于用于初始化 VLMO 的 BEIT 模型。ImageNet 的图像分辨率为 224×224，ADE20K 的图像分辨率为 512×512。我们对所有三个模型都在 ImageNet-21k 上进行了中间微调 [2]。

Models	ImageNet (acc@1)	ADE20K (mIoU)
ViT-Base	83.6	-
BEiT-Base	85.2	52.8
VLMo-Base	**85.5**	**53.4**

Table 3: Results on image classification and semantic segmentation.

Stagewise Pre-Training	NLVR2		Flickr30k	
	dev	test-P	TR	IR
Image-Only Pre-Training	80.33	81.06	95.60	87.69
Image-Only + Text-Only Pre-Training	**82.09**	**82.49**	**95.67**	**88.52**

Table 4: Ablation studies of stagewise pre-training, i.e., different initialization for vision-language pre-training. We report the average of R@1, R@5 and R@10 for Flickr30k. Results of NLVR2 are averaged over three runs.

4.6 Ablation Studies

Stagewise Pre-Training We first conduct ablation experiments of stagewise pre-training. ViLT [20] shows that using the ViT [12] model pretrained on image-only data as the initialization achieves better performance than the BERT model pretrained on text-only data. Therefore we start experiments with image-only pre-training. We compare using image-only pre-training, and image-only pre-training plus text-only pre-training as the initialization. For image-only pre-training, we directly use the parameters of BEiT-Base to initialize the self-attention module and all modality experts. For image-only pre-training plus text-only pre-training, we use pretrained parameters of BEiT-Base to initialize the vision expert and self-attention module of MoME Transformer, and then pretrain its language expert on text corpora. As shown in Table 4, image-only pre-training plus text-only pre-training improves our vision-language model. We also have tried to perform vision-language pre-training with random initialization but obtain a relatively low accuracy on downstream tasks. Stagewise pre-training effectively leverages large-scale image-only and text-only corpus, and improves our vision-language pre-training. Moreover, given the limited size of image-text pairs we used during pre-training, stage-wise pre-training on image-only and text-only data alleviates the need for image-text pair data.

MoME Transformer We also conduct ablation experiments of MoME Transformer. We employ ViT-Base to initialize the models for the ablation experiments. As present in Table 5, using MoME Transformer achieves better performance than standard Transformer for both retrieval and classification tasks. In addition, we also analyse the contribution of vision-language expert (VL-FFN) used in MoME Transformer. We remove the vision-language expert used in the top Transformer layers. Experimental results demonstrate that the introduction of vision-language expert improves the model. Using vision-language expert captures more modality interaction. Shared self-attention module used in MoME also positively contributes to our model. Section A presents the ablation study of shared self-attention module.

Pre-Training Tasks We perform ablation studies to analyse the contribution of different pre-training tasks, and the results are presented in Table 5. Compared with the model trained only using image-text contrastive loss, our unified training performs much better across classification and retrieval tasks. Introducing image-text matching with hard negative mining also greatly improves the model. This demonstrates the effectiveness of our unified-training framework with MoME Transformer. In addition, experimental results show that masked language modeling positively contribute to our model. Please refer to the supplementary material for more ablation studies.

Global Hard Negative Mining Different from ALBEF [23], which samples hard negatives from training examples of the single GPU (named as local hard negative mining). We perform hard negative mining from more candidates by gathering training examples of all GPUs (named as global hard negative mining). As shown in Table 6, our global hard negative mining brings significant improvements.

Models	ImageNet (acc@1)	ADE20K (mIoU)
ViT-Base	83.6	-
BEiT-Base	85.2	52.8
VLMo-Base	**85.5**	**53.4**

表 3：图像分类和语义分割的结果。

Stagewise Pre-Training	NLVR2		Flickr30k	
	dev	test-P	TR	IR
Image-Only Pre-Training	80.33	81.06	95.60	87.69
Image-Only + Text-Only Pre-Training	**82.09**	**82.49**	**95.67**	**88.52**

表 4：分阶段预训练的消融研究，即视觉-语言预训练的不同初始化方式。我们报告了 Flickr 30k 上 R@1、R@5 和 R@10 的平均值。NLVR2 的结果取三次运行的平均值。

4.6 消融研究

分阶段预训练 我们首先对分阶段预训练进行消融实验。ViLT [20] 表明，与使用在纯文本数据上预训练的 BERT 模型相比，使用在纯图像数据上预训练的 ViT [12] 模型进行初始化能够取得更好的性能。因此，我们从纯图像预训练开始实验。我们比较了以纯图像预训练，以及纯图像预训练加纯文本预训练作为初始化方式的效果。对于纯图像预训练，我们直接使用 BEIT-Base 的参数初始化自注意力模块和所有模态专家。对于纯图像预训练加纯文本预训练，我们使用 BEIT-Base 的预训练参数初始化 MOME Transformer 的视觉专家和自注意力模块，然后在文本语料库上对其语言专家进行预训练。如表 4 所示，纯图像预训练加纯文本预训练能够提升我们的视觉-语言模型性能。我们还尝试使用随机初始化进行视觉-语言预训练，但在下游任务上获得了相对较低的准确率。分阶段预训练能够有效利用大规模纯图像语料库和纯文本语料库，并提升我们的视觉-语言预训练效果。此外，鉴于我们在预训练期间使用的图像-文本对数量有限，在纯图像数据和纯文本数据上进行分阶段预训练减轻了对图像-文本对数据的需求。

MOME Transformer 我们还对 MOME Transformer 进行了消融实验。我们采用 ViT-Base 初始化用于消融实验的模型。如表 5 所示，在检索和分类任务中，使用 MOME Transformer 相比标准 Transformer 都能取得更好的性能。此外，我们还分析了 MOME Transformer 中使用的视觉-语言专家（VL-FFN）的贡献。我们移除了顶层 Transformer 层中使用的视觉-语言专家。实验结果表明，引入视觉-语言专家提升了模型性能。使用视觉-语言专家能够捕获更多模态交互。MOME 中使用的共享自注意力模块也对我们的模型有积极贡献。附录 A 给出了共享自注意力模块的消融研究。

预训练任务 我们通过消融研究分析不同预训练任务的作用，结果见表 5。与仅使用图文对比损失训练的模型相比，我们的统一训练方法在分类和检索任务上均表现出色。引入带难负样本挖掘的图文匹配任务，也显著提升了模型性能。这证明了采用 MOME Transformer 的统一训练框架是有效的。此外，实验结果表明，掩码语言建模对提升模型性能有积极作用。更多消融研究请参阅补充材料。

全局困难负样本挖掘 不同于 ALBEF [23] 从单个 GPU 的训练样本中采样困难负样本（称为局部困难负样本挖掘），我们通过汇聚所有 GPU 的训练样本，从更多候选样本中进行困难负样本挖掘（称为全局困难负样本挖掘）。如表 6 所示，我们的全局困难负样本挖掘带来了显著提升。

	Pre-Training Tasks			Transformer			NLVR2		Flickr30k	
	ITC	ITM	MLM	Std TRM	MoME	MoME−VLExp	dev	test-P	TR	IR
[1]	✓	✗	✗	✗	✓	✗	58.51	58.83	92.23	84.24
[2]	✓	✗	✓	✗	✓	✗	73.91	73.75	94.07	85.82
[3]	✓	✓	✗	✗	✓	✗	76.46	76.19	94.37	85.67
[4]	✓	✓	✓	✓	✗	✗	78.81	79.27	93.37	85.73
[5]	✓	✓	✓	✗	✗	✓	79.58	80.11	94.50	86.69
[6]	✓	✓	✓	✗	✓	✗	**80.13**	**80.31**	**95.17**	**87.25**

Table 5: Ablation studies of MoME Transformer and vision-language pre-training tasks. "ITC" is short for image-text contrastive loss, "ITM" is image-text matching, and "MLM" is masked language modeling. "Std TRM" is short for standard Transformer, and "MoME−VLExp" is MoME without VL experts. The average of R@1, R@5 and R@10 is reported for Flickr30k. Results of NLVR2 are averaged over three runs.

Models	NLVR2	
	dev	test-P
Local hard negative mining [23]	77.70	77.95
Global hard negative mining (ours)	**79.54**	**79.48**

Table 6: Global hard negative mining improves the model. We perform experiments using 32 V100 GPUs. The batch size per GPU is 32, and the total batch size is 1024. Local hard negative mining samples hard negatives from training examples of the single GPU (32 examples), while global hard negative mining uses training examples gathered from all GPUs as the candidates (1024 examples).

5 Conclusion

In this work, we propose a unified vision-language pretrained model VLMo, which jointly learns a dual encoder and a fusion encoder with a shared MoME Transformer backbone. MoME introduces a pool of modality experts to encode modality-specific information, and aligns different modalities using the shared self-attention module. The unified pre-training with MoME enables the model to be used as a dual encoder for efficient vision-language retrieval, or as a fusion encoder to model cross-modal interactions for classification tasks. We also show that stagewise pre-training that leverages large-scale image-only and text-only corpus greatly improves vision-language pre-training. Experimental results demonstrate that VLMo outperforms previous state-of-the-art models on various vision-language classification and retrieval benchmarks.

In the future, we would like to work on improving VLMo from the following perspectives:

- We will scale up the model size used in VLMo pre-training.
- We are also interested in fine-tuning VLMo for vision-language generation tasks, such as image captioning, following the method proposed in UniLM [11].
- We are going to explore to what extent vision-language pre-training can help each other modality, especially as the shared MoME backbone naturally blends in text and image representations.
- We can extend the proposed model to integrate more modalities (e.g., speech, video, and structured knowledge), supporting general-purpose multimodal pre-training.

References

[1] Hangbo Bao, Li Dong, Furu Wei, Wenhui Wang, Nan Yang, Xiaodong Liu, Yu Wang, Jianfeng Gao, Songhao Piao, Ming Zhou, and Hsiao-Wuen Hon. UniLMv2: Pseudo-masked language models for unified language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning, ICML 2020, 13-18 July 2020, Virtual Event*, volume 119 of *Proceedings of Machine Learning Research*, pages 642–652. PMLR, 2020. URL `http://proceedings.mlr.press/v119/bao20a.html`.

	Pre-Training Tasks			Transformer			NLVR2		Flickr30k	
	ITC	ITM	MLM	Std TRM	MoME	MoME−VLExp	dev	test-P	TR	IR
[1]	✓	✗	✗	✗	✓	✗	58.51	58.83	92.23	84.24
[2]	✓	✗	✓	✗	✓	✗	73.91	73.75	94.07	85.82
[3]	✓	✓	✗	✗	✓	✗	76.46	76.19	94.37	85.67
[4]	✓	✓	✓	✓	✗	✗	78.81	79.27	93.37	85.73
[5]	✓	✓	✓	✗	✗	✓	79.58	80.11	94.50	86.69
[6]	✓	✓	✓	✗	✓	✗	**80.13**	**80.31**	**95.17**	**87.25**

表 5：MOME Transformer 和视觉-语言预训练任务的消融实验。“ITC”是图像-文本对比损失（image-text contrastive loss）的缩写，“ITM”是图像-文本匹配（image-text matching）的缩写，“MLM”是掩码语言建模（masked language modeling）的缩写。“Std TRM”是标准 Transformer（standard Transformer）的缩写，“MoME−VLExp”表示不含视觉-语言专家的 MOME。Flickr30k 报告了 R@1、R@5 和 R@10 的平均值。NLVR2 的结果取三次运行的平均值。

Models	NLVR2	
	dev	test-P
Local hard negative mining [23]	77.70	77.95
Global hard negative mining (ours)	**79.54**	**79.48**

表 6：全局困难负样本挖掘可提升模型性能。我们使用 32 个 V100 GPU 进行实验。每个 GPU 的批量大小为 32，总批量大小为 1024。局部困难负样本挖掘从单个 GPU 的训练样本（32 个样本）中采样困难负样本，而全局困难负样本挖掘则使用从所有 GPU 汇总的训练样本作为候选样本（1024 个样本）。

5 结论

在这项工作中，我们提出了一种统一的视觉-语言预训练模型 VLMO，该模型利用共享的 MOME Transformer 主干网络联合学习双编码器和融合编码器。MOME 引入了一个模态专家池，用于编码模态特定信息，并通过共享的自注意力模块对齐不同模态。借助基于 MOME 的统一预训练，模型既可以作为双编码器，用于高效的视觉-语言检索，也可以作为融合编码器，对跨模态交互进行建模，以完成分类任务。我们还表明，利用大规模纯图像语料库和纯文本语料库进行分阶段预训练，能够显著提升视觉-语言预训练的效果。实验结果表明，在各种视觉-语言分类和检索基准测试中，VLMO 的性能优于此前的先进模型。

未来，我们希望从以下几个方面改进 VLMO：

- 我们将扩大 VLMO 预训练所使用的模型规模。
- 我们还对按照 UniLM [11] 中提出的方法，对 VLMO 进行微调以完成视觉语言生成任务（例如图像描述）感兴趣。
- 我们将探讨视觉-语言预训练在多大程度上能够帮助彼此的模态，尤其是在共享的 MOME 主干网络自然融合文本和图像表示的情况下。
- 我们可以扩展所提出的模型，以整合更多模态（例如语音、视频和结构化知识），从而支持通用多模态预训练。

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A Ablation Study of Shared Self-Attention

Table 7 presents the ablation study of shared self-attention module used in MoME Transformer for encoding image patches and text tokens. We compare shared self-attention with separate self-attention, which encodes image patches and text tokens using different attention parameters on the first L−F layers. The shared self-attention used in MoME achieves better performance. The shared self-attention module helps VLMo learn the alignment of different modalities, and fuse images and text at bottom layers for classification tasks.

Transformer	**NLVR2**		**Flickr30k**	
	dev	test-P	TR	IR
Separate Self-Attention	78.92	78.95	94.63	86.88
MoME (Shared Self-Attention)	**80.13**	**80.31**	**95.17**	**87.25**

Table 7: Ablation study of the shared self-attention module used in MoME. We experiment with separate attention on the first L−F layers, which encodes image patches and text tokens using different attention parameters.

B Hyperparameters for Text-Only Pre-Training

For the text-only pre-training data, we use English Wikipedia and BookCorpus [52]. AdamW [29] optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.98$ is used to train the models. The maximum sequence length is set to 196. The batch size is 1024, and the peak learning rate is 2e-4. We set the weight decay to 0.01. For the base-size model, we train the model for 500k steps. The large-size model is trained for 200k steps.

C Hyperparameters for Vision-Language Classification Fine-Tuning

Visual Question Answering (VQA) We fine-tune the models for 10 epochs with 128 batch size. The peak learning rate is 3e-5 for the base-size model, and 1.5e-5 for the large-size model. Following SimVLM [46], the input image resolution is 480×480. For VLMo-Large++, we use 768×768 image resolution.

Natural Language for Visual Reasoning (NLVR2) For results of Table 1, the models are fine-tuned for 10 epochs with 128 batch size. The peak learning rate of the base-size and large-size models are set to 5e-5 and 3e-5, respectively. The input image resolution is 384×384. For ablation experiments, we fine-tune the models for 10 epochs with 128 batch size, and choose learning rates from {5e-5, 1e-4}. The input image resolution is 224×224. All the ablation results of NLVR2 are averaged over 3 runs.

D Hyperparameters for Vision-Language Retrieval Fine-Tuning

COCO We fine-tune the base-size model for 20 epochs and large-size model for 10 epochs with 2048 batch size. The peak learning rate is 2e-5 for the base-size model and 1e-5 for the large-size model. The input image resolution is 384×384.

Flickr30K For results of Table 2, the base-size and large-size models are fine-tuned for 40 epochs with a batch size of 2048 and a peak learning rate of 1e-5. We use the fine-tuned model on COCO as the initialization. The input image resolution is 384×384. For all ablation experiments, we fine-tune the models for 10 epochs with 1024 batch size. The peak learning rate is set to 5e-5, and the input image resolution is 224×224.

共享自注意力机制的消融研究

表 7 展示了 MOME Transformer 中用于编码图像块和文本标记的共享自注意力模块的消融研究。我们将共享自注意力与独立自注意力进行了比较，后者在前 L−F 层使用不同的注意力参数分别编码图像块和文本标记。MOME 中采用的共享自注意力取得了更好的性能。共享自注意力模块有助于 VLMO 学习不同模态之间的对齐关系，并在底层融合图像和文本，以完成分类任务。

Transformer	**NLVR2**		**Flickr30k**	
	dev	test-P	TR	IR
Separate Self-Attention	78.92	78.95	94.63	86.88
MoME (Shared Self-Attention)	**80.13**	**80.31**	**95.17**	**87.25**

表 7：MOME 中使用的共享自注意力模块的消融实验。我们在前 L−F 层采用独立注意力进行实验，使用不同的注意力参数对图像块和文本标记进行编码。

B 纯文本预训练的超参数

对于纯文本预训练数据，我们使用英文维基百科和 BookCorpus [52]。使用 AdamW [29] 优化器，并将 $\beta_1 =$ 设置为 0.9、将 $\beta_2 =$ 设置为 0.98，以训练模型。最大序列长度设置为 196。批量大小为 1024，峰值学习率为 2e-4。我们将权重衰减设置为 0.01。对于基础规模模型，我们训练模型 50 万步。大规模模型训练 20 万步。

C 视觉-语言分类微调的超参数

视觉问答（VQA） 我们使用批大小为 128 的设置对模型进行 10 个 epoch 的微调。基础尺寸模型的峰值学习率为 3e-5，大尺寸模型的峰值学习率为 1.5e-5。遵循 SimVLM [46]，输入图像分辨率为 480 × 480。对于 VLMO-Large++，我们使用 768 × 768 的图像分辨率。

视觉推理自然语言（NLVR2）对于表 1 的结果，模型以 128 的批量大小微调 10 个 epoch。基础尺寸和大尺寸模型的峰值学习率分别设置为 5e-5 和 3e-5。输入图像分辨率为 384 × 384。对于消融实验，我们以 128 的批量大小对模型微调 10 个 epoch，并从 {5e-5, 1e-4} 中选择学习率。输入图像分辨率为 224 × 224。NLVR2 的所有消融结果均为 3 次运行的平均值。

D 视觉-语言检索微调超参数

COCO 我们使用 2048 的批量大小，对基础尺寸模型微调 20 个 epoch，对大型尺寸模型微调 10 个 epoch。基础尺寸模型的峰值学习率为 2e-5，大型尺寸模型的峰值学习率为 1e-5。输入图像分辨率为 384 × 384。

Flickr30K 对于表 2 的结果，base-size 和 large-size 模型均以 2048 的批量大小和 1e-5 的峰值学习率进行 40 个 epoch 的微调。我们使用在 COCO 上微调后的模型作为初始化模型。输入图像分辨率为 384 × 384。对于所有消融实验，我们使用 1024 的批量大小对模型进行 10 个 epoch 的微调。峰值学习率设置为 5e-5，输入图像分辨率为 224 × 224。